Arista Access Node MAR-ACC-45 Major QoS Congestion Failure (ALARM_4109) Resolution SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_4109

Description

The failure domain in question pertains to Quality of Service (QoS) congestion events on the Arista Access Node MAR-ACC-45, a crucial component of the Marathahalli Core site's infrastructure. QoS congestion events, characterized by microbursts, interface tail drops, and QoS queue starvation, are typically precipitated by excessive traffic volume exceeding the configured burst and committed information rates (BIR and CIR) or inadequate buffer memory allocation. The Arista Access Node, operating under the egress QoS policy, scrutinizes traffic flows against pre-configured policing and shaping rules, effectively regulating bandwidth allocation and traffic prioritization. However, when the traffic volume exceeds the node's capacity, microbursts and tail drops emerge, while QoS queue starvation ensues due to inadequate buffer memory allocation. Consequently, this necessitates the implementation of specific diagnostic, resolution, and verification procedures to ascertain the root cause and rectify the situation.

Pre-requisite Checks

- Confirm the site location and node ID (MAR-ACC-45) to ensure the correct device is being addressed.

- Verify the current CLI session is active and logged in with administrator privileges.

- Ensure remote hands are dispatched to Marathahalli Core, if necessary, to perform physical checks or repairs.

- Verify redundant link status and ensure it is operational and not contributing to the congestion event.

Potential Root Causes

Cause	Probability
Inadequate buffer memory allocation	High
Excessive traffic volume exceeding BIR and CIR	Medium
Misconfigured egress QoS policy	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface ethernet 1/1 counters, Expected Output: Interface statistics, including errors, drops, and packets transmitted., Description: This command isolates interface-level traffic metrics, aiding in the identification of potential congestion hotspots.

Command: show qos policer, Expected Output: List of configured policers, including their rates and burst sizes., Description: This
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command scrutinizes the configured policers, enabling the assessment of their adequacy in regulating traffic flow.

Command: show buffer memory allocation, Expected Output: Current buffer memory allocation status, including utilized and free memory., Description: This command assesses the current buffer memory allocation status, facilitating the determination of potential buffer memory exhaustion.
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Resolution Steps

Step 1: Step: Adjust Buffer Memory Allocation, Command: buffer memory allocation , Description: Modify buffer memory allocation to alleviate potential buffer memory exhaustion.

Step 2: Step: Reconfigure Egress QoS Policy, Command: configure terminal egress qos policy policing cir pir , Description: Amend the egress QoS policy to ensure it adequately regulates traffic flow.

Step 3: Step: Increase Interface Buffer Size, Command: interface ethernet 1/1 buffer-size , Description: Augment the interface buffer size to reduce the likelihood of tail drops.

Post-Check Verification

```
Command: show interface ethernet 1/1 counters, Expected Output: Reduced error and drop rates, along with increased packets transmitted., Description: Verify interface-level traffic metrics have improved, indicating the resolution has been effective.

Command: show qos policer, Expected Output: Updated policer configurations, reflecting the adjustments made., Description: Verify the configured policers have been adjusted to adequately regulate traffic flow.

Command: show buffer memory allocation, Expected Output: Improved buffer memory allocation status, with increased free memory., Description: Verify the buffer memory allocation has been adjusted to alleviate potential buffer memory exhaustion.
```

Roll-back Procedures

- Step: Restore Original Buffer Memory Allocation, Command: buffer memory allocation , Description: Revert buffer memory allocation to its original state.

- Step: Restore Original Egress QoS Policy, Command: configure terminal egress qos policy policing cir pir , Description: Revert the egress QoS policy to its original configuration.

- Step: Restore Original Interface Buffer Size, Command: interface ethernet 1/1 buffer-size , Description: Revert the interface buffer size to its original value.